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Course: Software Design and Architecture
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Lab Task:

Food Delivery System

A food delivery system allows customers to browse restaurant, select food items, place order, make payments, and receive deliveries.

Conceptual class:

1. Customer (A person who orders the food through application)
2. Restaurant (A business that prepares and sells food)
3. Menu (Contains the food items offered by a restaurant)
4. Food item (Represent an individual food item available for ordering)
5. Order (Represents a customer's food order)
6. Payment (Records payment information and payment status)
7. DeliveryDriver (A person responsible for delivering food)
8. Delivery (Represents the process of delivering an order)
9. Review (Stores customer feedback and ratings)
10. Address (Represents location where an order should be deliver)

Attributes and Methods of 3 important classes:

Class 1: Customer

Attributes:

customerID - Unique identification number of customer.
name - Name of the customer.
phoneNumber - contact number of the customer
email - Email address of customer.
address - Delivery address of customer.

Methods:

Register() - Register a new customer
Login() - Allows customer to access the system
placeOrder() - Place a new food order
updateProfile() - Updates customer information.
viewOrderHistory() - Displays previous order.

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Class 2: Order

Attributes:

orderID- Unique identification number of order.

orderDate- Date and time when the order was placed.

totalAmount-Total cost of the order.

orderStatus-current status of the order.

customerID- Identifies the customer who place the order.

Methods:

createOrder()- Creates a new food order.

calculateTotal()-Calculates the total cost of the order.

cancelOrder()-Cancel an order when cancellation is allowed.

updateStatus()-Update the order status.

viewOrderDetails()-Displays the details of an order.

Class 3: DeliveryDriver

Attributes:

driverID- Unique identification number of driver.

name- Name of delivery driver.

phoneNumber-Contact number of the driver.

vehicleNumber-Registration number of vehicle.

availabilityStatus- Indicates whether the driver is available.

Methods:

acceptDelivery()-Accept a delivery assignment.

updateLocation()-Updates the driver's current location.

pickUpOrder()-Collects the prepared order from restaurant.

deliveOrder()- Delivers the order to the customer.

updateAvailability()- Changes the driver's availability status.

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Abstraction Decision:

What real world details did you deliberately ignore, why?

In the Food Delivery System, I focused only on the information and behaviors necessary for ordering and delivering food. I ignored annecessary details such as the customer's physical appearance, the resturant's interior design and the driver's personal hobbies because they do not affect the ordering process. I included important information such as customer details, food items, orders, payments and delivery status. This abstraction makes the software system simpler, easier to understand and easier to maintain.

Invented Software Object:

OrderYaliator: It is a software object that checks whether an order satisfies the system's requirements before it is confirmed.

Attributes:

validationStatus, errorMessage.

Methods:

validateOrder(), checkAvailability(), checkDeliveryAddress()
getValidationResult(),

Why is it an invented software object?

Order validator does not have a direct physical counter part in the real world. It is a software object created to perform validation task automatically. It helps prevent in complete our invalid orders from being confirmed. This improves the reliabity and efficiency of the food delivery system.

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